UTTARAKHAND

Food Heritage and Agriculture

A research study documenting Uttarakhand’s traditional foods, regional recipes, indigenous crops, seasonal practices and mountain agriculture.

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Research Report

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Welcome to Pahari Food

The cooking and farming of the Uttarakhand Himalaya

Uttarakhand’s food is the cooking of a mountain people: honest, seasonal and built from what the hills themselves provide. Grown on steep terraces, most of it rain-fed, the staples are millets, hardy pulses, wild greens and fruit, brought together with ghee, curd and a splash of mustard oil. There are very few heavy masalas. Flavour comes from smoke, from slow cooking in iron vessels, and from a few signature seeds and salts that exist nowhere else in Indian cooking.

The state is divided into two cultural kitchens. Garhwal (Chamoli, Tehri, Pauri, Uttarkashi, Dehradun, Haridwar and Rudraprayag) and Kumaon (Nainital, Almora, Pithoragarh, Udham Singh Nagar, Champawat and Bageshwar) share many dishes under the common name Pahari cuisine, yet each has its own favourites. Around them sit the distinct food worlds of the Jaunsari, Bhotiya, Tharu and Buksa communities.

This encyclopedia walks through that world one topic at a time: the grains and pulses, the seasonings, the fruits and drinks, the dishes, the farming that produces them and the old methods that kept food safe through long Himalayan winters.

Uttarakhand at a glance

Districts	13 (7 in Garhwal, 6 in Kumaon)
Hill districts	9 of 13; Nainital, Haridwar, Dehradun and Udham Singh Nagar also have plains
Irrigated farmland	Only about 10–13%, so most farming is rain-fed
Signature crops	Mandua, jhangora, gahat, bhatt, rajma, chaulai, red rice, basmati
Signature seasoning	Jakhya seeds fried in mustard oil
Traditional farming system	Barahnaja, “twelve seeds” grown together in one field
GI-tagged products	27 by December 2023; more than 30 by July 2026
Main communities	Garhwali, Kumaoni, Jaunsari, Bhotiya, Tharu, Buksa, Raji

Grains and Millets

The cereals that feed the hills

Mandua

Finger millet · Koda (Garhwali) · Ragi

Mandua is the backbone of hill food. Its dark flour makes thick rotis, porridge and dumplings that give energy through cold winters and hard farm work. The plant tolerates drought and poor soil, which is why it has fed Himalayan villages for generations. It is regarded as the lead crop of the Barahnaja system.

> **Did you know** Mandua received a Geographical Indication (GI) tag in 2023.

Jhangora

Barnyard millet · Jhangore

A small, pale grain grown in rain-fed Himalayan fields. It is cooked like rice, simmered into porridge and made into the beloved jhangore ki kheer, a creamy milk pudding that traditionally ends a Garhwali meal.

> **Did you know** Jhangora is GI-tagged and is cooked much like rice.

Kauni and Cheena

Foxtail millet and proso millet

Two lesser-known millets that sit alongside mandua and jhangora in the hill farmer’s field. Farmer and seed conservationist Vijay Jardhari calls these millets the lifeline of hill people. They are cooked as porridge or rice-like dishes and store well.

> **Did you know** Both are drought-tolerant and were once pushed aside by hybrid seeds and cash crops.

Chaulai

Amaranth · Ramdana

Chaulai gives two foods in one: tender leaves eaten as a green, and tiny seeds that are eaten during fasts and popped for sweets. Its red and golden seed heads are a familiar sight in autumn fields.

> **Did you know** GI-tagged in 2023 as Uttarakhand Chaulai (Ramdana).

Kuttu

Buckwheat

Not a true grain but a seed, kuttu is grown at higher altitudes. Its flour makes the flatbreads eaten on fasting days, and its leaves are also cooked as a vegetable.

> **Did you know** Kuttu is gluten-free despite the name “buckwheat”.

Red rice

Lal chawal

A traditional hill rice with a reddish bran and nutty flavour, grown on irrigated terraces. It is cooked plain with dal or used in khichdi and kheer.

> **Did you know** Uttarakhand red rice carries a GI tag.

Doon Basmati

Basmati of Dehradun valley

The fragrant long-grain rice grown in the plains and valley of Dehradun is part of the multi-state Basmati Geographical Indication that Uttarakhand shares with other northern states.

> **Did you know** Basmati was one of the first nine Uttarakhand products to receive a GI tag.

Wheat

Gehun

Wheat is the winter crop of the hills, used for rotis and parathas. In Jaunsar-Bawar it is also fermented into the soft, griddle-cooked crepe called chilra.

> **Did you know** Fermented wheat batter is a hallmark of Jaunsar-Bawar cooking.

Pulses and Legumes

Protein from the terraces

Gahat

Horse gram · Kulath

The pulse of the dry hills and one of the most distinctive tastes of the region. Gahat is cooked into dal, ground into batter for dubuk and phaanu, and rolled into stuffed parathas. Its warming quality makes it a winter favourite.

> **Did you know** GI-tagged in 2023. Village tradition links it with kidney-stone prevention, which is folk belief, not medical advice.

Bhatt

Black soybean · Kala bhat

A glossy black soybean that is high in protein and cooks into a rich, earthy dal. It is made into bhat ki dal, into the thick bhatt ki churdkani, and sometimes roasted as a snack.

> **Did you know** Uttarakhand Kala Bhatt holds a GI tag.

Munsiyari Rajma

White kidney beans

Grown in the high valleys around Munsiyari, these creamy white beans are prized for soft texture and quick cooking compared with ordinary rajma.

> **Did you know** Munsiyari white rajma was among the first nine GI products of the state.

Pahari Toor

Hill pigeon pea · Arhar

A local toor (arhar) dal grown in the hills and valued for its flavour. It is cooked plain or ground into pastes such as dubuk.

> **Did you know** GI-tagged as Pahari Toor in 2023.

Urad

Black gram

Urad is the multipurpose pulse. It is roasted for chainsoo, ground and sun-dried into badi nuggets, and fermented during sun-drying to make long-keeping dried foods.

> **Did you know** The urad paste in sun-dried nuggets begins to ferment within three to four hours of drying.

Spices and Seasonings

Small ingredients, big identity

Jakhya

Wild mustard-type seed

The seasoning that defines Pahari cooking. Tiny dark seeds are dropped into hot mustard oil, where they crackle and release a nutty, mildly pungent aroma. A pinch transforms a plain dal or potato dish and uses little fuel.

> **Did you know** Jakhya is one of the most recognised flavours of Pahari cooking.

Tejpat

Indian bay leaf

The aromatic leaf used across Indian kitchens is gathered from trees in Uttarakhand's forests. It flavours rice, curries and spice blends.

> **Did you know** Tejpat was the first botanical plant from Uttarakhand to receive a GI tag.

Lakhori Mirch

Almora chilli

A chilli of the Almora hills, valued for its colour and heat. It is dried and used in chutneys and tempering.

> **Did you know** GI-tagged as Almora Lakhori Chilli.

Pisyu Loon

Flavoured hill salt

A coarse, hand-ground salt blend of Kumaon, traditionally pounded on a stone with garlic, chilli and herbs. It is eaten with fruit such as cucumber and citrus, and with rotis.

> **Did you know** Each family has its own blend.

Bhang Seeds

Hemp seeds

The culinary seeds of the hemp plant are dry-roasted and ground into bhang ki chutney, a tangy, nutty relish that is one of Kumaon's best-loved side dishes. Only the seeds are used.

> **Did you know** Use food-grade hemp seeds from a legal commercial source.

Timur

Himalayan Sichuan pepper

A wild, citrusy peppercorn that gives a slightly tingling sensation. It is used in chutneys and seasonings and is collected from the hills.

> **Did you know** It is related to Sichuan pepper.

Faran / Jambu

Himalayan chive-like herb

A mountain herb that is dried and fried in oil to give a garlicky, onion-like tempering to dals and vegetables.

> **Did you know** It grows wild at high altitude and is collected rather than farmed.

Galgal

Hill lemon

A large, thick-skinned, fragrant citrus used for pickles and for the sharp, cooling nimbu saan, crushed with salt, sugar or curd.

> **Did you know** Its juice gives chutneys a sharp, fresh tang.

Fruits of the Hills

Orchard crops and wild berries

Ramgarh Peach

Aadu (Kumaoni)

A juicy summer peach from the orchards of Ramgarh in Nainital district.

> **Did you know** GI-tagged in 2023.

Ramnagar Litchi

Nainital litchi

A fragrant litchi grown in the warm foothill belt near Ramnagar and ripening in early summer.

> **Did you know** GI-tagged in 2023.

Malta

Hill citrus

A sweet-tangy citrus that ripens in winter, widely sold in hill markets.

> **Did you know** Uttarakhand Malta has a GI tag.

Bedu

Himalayan fig

A small wild fig that grows on hillsides and ripens in summer, eaten fresh and dried.

> **Did you know** Bedu is displayed among the state’s agricultural GI products at the Haldwani gallery.

Kafal

Bayberry

A tart-sweet red berry that ripens in late spring and is sold in small baskets beside mountain roads, often with a sprinkle of salt and chilli.

> **Did you know** Kafal is celebrated in Kumaoni folk songs.

Hisalu and Kilmora

Himalayan raspberry and barberry

Wild berries gathered by children and villagers. Hisalu is golden and sweet-sour; kilmora is small, purple and tangy.

> **Did you know** Both grow wild and rarely reach cities.

Orchard fruits

Apple · Pear · Plum · Apricot · Walnut

Higher valleys grow apples, pears, plums, apricots (chuli) and walnuts, which are eaten fresh, dried or pressed for oil.

> **Did you know** Fruit is also dried and stored for winter.

Drinks and Beverages

Cooling, warming and fermented

Buransh Sharbat

Rhododendron juice

Made from the deep red flowers of the rhododendron tree (Rhododendron arboreum), buransh sharbat is a sweet-sour, cooling drink prepared in spring when the hills turn crimson.

> **Did you know** GI-tagged in 2023. The flower is also the state tree’s blossom.

Berinag Tea

Kumaon hill tea

Tea grown in the Berinag area of Pithoragarh district, harvested in the cool mountain air.

> **Did you know** GI-tagged in 2023.

Mattha

Spiced buttermilk

Whey or buttermilk, served salted and sometimes tempered, drunk with meals to cool the body.

> **Did you know** Dairy forms of curd, mattha and ghee are central to hill food.

Lugdi

Traditional fermented drink

A traditional drink of the Jaunsar-Bawar region produced by fermentation, valued locally for warmth in the cold.

> **Did you know** It is part of a wider tradition of fermented foods in the Tons valley.

Chhang / Jaad

High-valley fermented beer

A mild barley or millet drink made in Bhotiya high-altitude communities.

> **Did you know** Local practice varies widely between villages.

Traditional Dishes

Everyday meals of Garhwal and Kumaon

Kafuli

Kumaoni and Garhwali · Winter

A thick, glossy green curry of spinach and fenugreek simmered with ground rice and finished with ghee. Cooked slowly in an iron kadhai, it is one of the most iron-rich, comforting dishes of Uttarakhand and is eaten with rice or mandua roti.

Did you know One of the best-known dishes of Uttarakhand.

Phaanu

Garhwali · Winter

A stew-like dal made from several pulses soaked overnight, ground and cooked with very little spice. Its thick, porridge-like body reflects a kitchen that makes the most of dried stores in winter.

Did you know The pulse mix changes with whatever the household has.

Chainsoo

Garhwali · Winter

Whole black urad is dry-roasted until fragrant, ground coarsely and simmered in an iron pan. The roasting gives a deep, nutty flavour and the dish is easy to digest.

Did you know Spellings include chainsu, chaunsa and chainsi.

Bhat ki Dal and Churdkani

Kumaoni

Black soybean dal is cooked slowly with minimal spices, while bhatt ki churdkani is a thicker preparation that shows the farming-based diet at its best.

Did you know Black soybean is a traditional crop of hill terraces.

Dubuk

Kumaoni

Arhar, gahat or another lentil is ground into a fine paste with a little rice flour and cooked until thick, eaten hot with rice and ghee.

Did you know Its texture sits between a dal and a porridge.

Baadi

Garhwali

Mandua flour is stirred into hot water into a firm dough ball and served with ghee or a runny dal. It is a strength food for farm work.

Did you know Often paired with gahat dal in winter.

Aloo ke Gutke

Kumaoni

Boiled potatoes are broken up and tossed in hot mustard oil with jakhya, red chilli, turmeric and coriander until the edges crisp. It is a favourite with puri and on travel days.

Did you know Gutke means small chunks.

Kandali ka Saag

Garhwali

Himalayan nettle leaves are boiled to remove the sting and cooked with simple spices. Nutritious and foraged rather than farmed, it shows how mountain cooks turn wild plants into food.

Did you know Cooking removes the sting completely.

Badi ki Sabji

Kumaoni

Sun-dried lentil nuggets (badi) are fried and simmered into a curry, often without onion or tomato, and served with plain rice.

Did you know Making badi in summer for winter use is a village tradition.

Bhang ki Chutney

Kumaoni and Garhwali

Roasted hemp seeds are ground with green chilli, salt and lemon into a sharp, nutty chutney. It is eaten with rotis, boiled potatoes and festival meals.

Did you know Eaten especially around Holi.

Kumaoni Raita

Kumaoni · Summer

Grated cucumber, preferably the large ripe pahadi kheera, is mixed with curd and a paste of mustard seeds, turmeric and salt. It is yellow, cooling and a little fiery.

Did you know The pulp of ripe pahadi cucumber is also sun-dried into nuggets.

Thechwani

Kumaoni

A rustic dish made of crushed vegetables such as radish or potato and tempered with local seasonings.

Did you know Thechna means to crush.

Sweets and Festival Foods

Treats made for weddings, journeys and festivals

Arsa

Garhwali and Kumaoni

Rice flour is kneaded with jaggery and fennel or poppy seeds, shaped into small discs and fried in ghee or oil. Crisp outside and soft inside, it appears at weddings and festivals.

Did you know Arsa keeps well, so it is a classic gift sweet.

Bal Mithai

Almora

Brown roasted khoya fudge coated in tiny sugar balls. Almora's most famous sweet is sold in packed boxes at every hill station.

Did you know Almora is known as the home of bal mithai.

Singori

Almora

A khoya sweet wrapped in a leaf of the maalu creeper, which gives it a faint, fresh aroma.

Did you know The leaf is not eaten; it only flavours the sweet.

Khajure

Kumaoni

Fried dough biscuits sweetened with jaggery, made as gifts when someone travels to relatives because they stay fresh for days.

Did you know In old villages the smell of khajure meant a family member was leaving.

Ghughuti

Kumaoni · Makar Sankranti

Fried sweet dough in shapes like birds, drums and flowers, strung on garlands for Uttarayani. Children wear the garlands and feed crows with them.

Did you know The festival is also called Ghughutiya.

Jhangore ki Kheer

Garhwali

Barnyard millet slowly simmered in milk with cardamom and jaggery or sugar. Rich, mildly nutty and served warm.

Did you know A GI-tagged grain turned into a dessert.

Foods of the Mountain Communities

Jaunsari, Bhotiya, Tharu, Buksa and Raji

Jaunsari (Jaunsar-Bawar)

Dehradun and Tehri hills

The state’s largest tribal community has a distinct food culture built on wheat, fermentation and local drinks. Foods recorded include balam, jaan, kacchi, sez, bhatooru, aske, chilra, dhindki, kadhiyiek, siddu, seera, mashyada bhaat, sattu, pakhli sur and baari. Chilra is a fermented wheat crepe served with ghee or honey, and siddu is a steamed fermented bread of the Tons valley.

Bhotiya

High valleys of Chamoli, Pithoragarh, Uttarkashi

Food shaped by high altitude and old links with Tibet. These communities were traditional traders and herders, and their dishes lean on hardy grains, dairy and dried foods. Detailed documentation is still limited.

Tharu and Buksa

Terai of Udham Singh Nagar

The Tharu are the second-largest tribal community and live in the Terai. Their cooking centres on rice and freshwater foods, and rice-flour steamed rolls (dhikri) are a known Tharu dish across the Terai. The Buksa live mainly around Gadarpur, Ramnagar, Bajpur and Kashipur and are classed as a Particularly Vulnerable Tribal Group.

Raji

Pithoragarh area

The smallest tribal group of the state; its food traditions are very little documented.

Seasonal Food Calendar

What the hills eat through the year

Season	Foods and activities
Winter (Nov–Feb)	Kafuli, phaanu, chainsoo, gahat and bhatt dishes, mandua roti, baadi, malta, jaggery and ghee. Ghughuti on Makar Sankranti.
Spring (Mar–Apr)	Rhododendron flowers bloom and buransh sharbat is made. Fresh greens such as nettle and wild leaves reach kitchens.
Summer (May–Jun)	Kafal, hisalu and kilmora ripen. Peaches and litchi arrive from Nainital district. Cucumber raita, badi and sun-dried foods are prepared.
Monsoon (Jul–Sep)	Mixed Barahnaja crops grow in the fields. Pahadi cucumber, bhang ki chutney and gulthia are enjoyed.
Autumn (Oct–Nov)	Millet harvest of mandua, jhangora and chaulai. Apples and walnuts. Families store grain, dry vegetables and prepare for winter.

Timings are approximate and shift with altitude and weather.

District Guide

A quick look at 13 districts

District	Known for
Dehradun	Doon basmati; the Jaunsar-Bawar fermented foods of Chakrata.
Haridwar	Plains district; wheat, rice and sugarcane products.
Tehri Garhwal	Jardhar village is known for Barahnaja farming; widely grown jhangora and mandua.
Uttarkashi	Jaunsari and Bhotiya foods; apple and rajma in high valleys.
Chamoli	Bhotiya foods and potato and pulse farming at altitude.
Rudraprayag	Hill staples; local specialities still to be documented.
Pauri Garhwal	Hill staples; local specialities still to be documented.
Nainital	Ramgarh peach and Ramnagar litchi; Haldwani houses the state GI gallery.
Almora	Bal mithai, singori and Lakhori chilli.
Pithoragarh	Munsiyari white rajma, Berinag tea and the largest Bhotia population in Kumaon.
Bageshwar	Hill staples and Bhotia communities; dishes still to be documented.
Champawat	Tharu and hill communities; dishes still to be documented.
Udham Singh Nagar	Tharu and Buksa food culture of the Terai; rice, wheat and sugarcane belt.

Traditional Farming

Growing food the Himalayan way

Barahnaja: the twelve-seed system

Barahnaja means “twelve seeds”. It is an ancient Garhwali method of growing around twelve crops together in one field: millets, pulses, oilseeds and greens sown at the start of the monsoon. Mandua is regarded as the head of the system, and amaranth is another key crop. The crops support each other: pulses put nutrients back into the soil, tall and short plants share space, and if one crop fails the others survive. No pesticides or fertilisers are used, so the system is called organic by default. In Jardhar village of Tehri district, about 375 farmers still practise it, and a family of five can be fed for a year from two acres or less.

The Saar system

Unirrigated land is split into two cropping periods. One set of fields carries mandua and Barahnaja, the other carries jhangora and unirrigated paddy. This keeps food diverse and the soil fertile year after year.

Terraces and livestock

Hill farmers cut narrow terraces into steep slopes. These hold soil and water, and they also grow fruit, fodder and medicinal plants. Almost every family keeps a buffalo or cow, whose milk feeds the household and whose manure feeds the fields.

Seed saving

Traditional seeds are passed from household to household. The Beej Bachao Andolan (Save Seed Movement), led by Vijay Jardhari, works to conserve traditional varieties that hybrid seeds and monoculture pushed aside.

Organic Uttarakhand

The Uttarakhand Organic Commodity Board was formed in 2003 to support organic farming, and the Uttarakhand Organic Products Council now helps producers register GI products.

Traditional Food Preservation

How hill households kept food through winter

Method	How it works
Sun-drying	Lentil nuggets (badi) and sun-dried vegetables are made in summer for the winter. Villagers mix grated ripe pahadi cucumber with urad paste and spices, then dry the nuggets in the sun.
Pickling	Salt, oil and spices keep fruit and vegetables for months. Galgal, chilli and radish pickles are typical, and jars are often left in the sun each morning.
Fermentation	Chilra, siddu and aske in Jaunsar-Bawar are fermented foods, and lugdi is a fermented drink.
Dairy	Ghee, curd and mattha are the main sources of fat and give the food a long shelf life.
Drying and pressing	Foods are dried or pressed so they last through winter scarcity.
Durable sweets	Khajure and arsa are made partly because they stay fresh for days, ideal for travel.

GI-Tagged Foods

Products with official Geographical Indication status

A Geographical Indication (GI) tag certifies that a product comes from a specific place and carries its qualities. These food and agriculture products from Uttarakhand have received GI status:

- Basmati rice (shared multi-state tag)
- Tejpat (Indian bay leaf)
- Munsiyari white rajma
- Kachyura (chyura) oil
- Mandua
- Jhangora
- Gahat
- Red rice
- Kala Bhatt
- Chaulai (Ramdana)
- Almora Lakhori chilli
- Pahari Toor dal
- Buransh sharbat
- Berinag tea
- Malta
- Ramnagar litchi
- Ramgarh peach

Fascinating Facts

Quick things worth knowing

- Uttarakhand received 18 GI certificates on a single day, 4 December 2023, a record among Indian states at that time. That brought its total to 27.
- Tejpat (Indian bay leaf) was the first botanical plant in the state to get a GI tag.
- Barahnaja literally means “twelve seeds”, and one two-acre farm can feed a family of five for a year.
- Mandua is called Koda in Garhwali and is treated as the head of the Barahnaja system.
- Only about 10 to 13% of the state’s farmland is irrigated, so millets and pulses rule the hill kitchen.
- A signature Pahari tempering uses jakhya seeds rather than cumin.
- The urad paste in sun-dried nuggets begins to ferment within three to four hours of sunning.
- Khajure was traditionally made the evening before a journey to visit relatives.
- Bal mithai and singori are specialities of Almora.
- The state government has estimated that more than 100 local products could qualify for GI tags.
- In July 2026 Uttarakhand opened its first dedicated GI products gallery at the Forest Training Academy, Haldwani.
- The Jaunsari are the state’s largest tribal community; the Buksa are a Particularly Vulnerable Tribal Group.

Glossary

Local names and what they mean

Aadu	Peach (Kumaoni)	**Jhangora**	Barnyard millet
Arsa	Fried rice-jaggery sweet	**Kafal**	Bayberry
Badi	Sun-dried lentil nugget	**Koda**	Garhwali name for mandua
Bhatt	Black soybean	**Kuttu**	Buckwheat
Buransh	Rhododendron	**Mandua**	Finger millet
Chaulai	Amaranth	**Mattha**	Buttermilk
Gahat	Horse gram	**Pisyu loon**	Hill salt blend
Galgal	Hill lemon	**Ramdana**	Amaranth seeds
Jakhya	Tempering seed	**Tejpat**	Indian bay leaf

Still to Be Documented

Topics that need more local research

Topic	What is needed
District details	Specific food traditions of Pauri Garhwal, Rudraprayag, Bageshwar and Champawat need local interviews.
Barahnaja crop list	The exact twelve crops differ by village. Collect the list from farmers in each area.
Bhotiya and Raji cooking	Very little written material exists. Documenting dishes needs fieldwork.
Recipe measurements	Home cooks measure by hand. Recipes in this encyclopedia are starting points and should be test-cooked before publishing.
Nutrition data	Add lab-tested nutrient values for mandua, gahat and bhatt before making health claims.
Local-language names	Add Garhwali, Kumaoni and Jaunsari names for each ingredient and dish.

Cooking the Pahari Way

A few notes before you begin

Traditional hill cooking is simple. It uses mustard oil, jakhya seeds, iron vessels and slow heat, with very little masala. Heat the mustard oil until it just smokes, then let it cool a moment before adding seeds. This mellows the raw sharpness.

Home cooks measure by hand and by eye. Treat the quantities here as starting points and adjust to your taste. Jakhya can be replaced with cumin if it is not available.

Kafuli

Spinach and fenugreek green curry · Serves 4 · 35 minutes

Ingredients

- 500 g spinach
- 150 g fresh fenugreek leaves
- 2 tbsp rice flour
- 2 tbsp mustard oil
- 1 tsp jakhya or cumin
- 2 dried red chillies
- 4 garlic cloves, chopped
- 1 tsp grated ginger
- ¼ tsp turmeric
- Salt
- 1 tsp ghee

Method

1. Blanch the greens for 2 minutes, drain and grind coarsely.
2. Heat the oil, add jakhya, chillies, garlic and ginger.
3. Add turmeric and the green puree.
4. Whisk the rice flour into 1 cup of water and stir it in.
5. Simmer 15 to 20 minutes, stirring, until thick and glossy. Add salt.
6. Top with ghee. Serve with rice or mandua roti.

> **Tip** Cook in an iron kadhai for the traditional flavour.

Phaanu

Mixed-lentil stew · Serves 4 · 30 minutes plus soaking

Ingredients

- 1 cup mixed pulses (gahat, urad, moong, masoor, toor)
- 2 tbsp mustard oil
- 1 tsp jakhya
- Pinch of asafoetida
- 1 tsp ginger
- 1–2 green chillies
- ¼ tsp turmeric
- Salt
- About 2 cups water

Method

1. Soak the pulses overnight, drain and grind coarsely with very little water.
2. Temper the oil with jakhya, asafoetida, ginger and chilli.
3. Add the paste and turmeric and stir on low heat.
4. Add hot water as needed and cook 15 to 20 minutes into a thick stew.
5. Stir constantly so it does not catch. Serve with rice and ghee.

> **Tip** Use whichever pulses you have; the mix changes the flavour.

Chainsoo

Roasted black-lentil dal · Serves 4 · 40 minutes

Ingredients

- 1 cup whole black urad
- 2 tbsp mustard oil
- 1 tsp jakhya or cumin
- 2 garlic cloves
- 2 dried red chillies
- 3 cups water
- Salt

Method

1. Dry roast the dal on low heat for 8 to 10 minutes until fragrant and darker.
2. Cool and grind coarsely.
3. Heat oil, temper with jakhya, garlic and chillies.
4. Add the ground dal, sauté 2 minutes, then add water gradually.
5. Simmer 25 to 30 minutes, stirring. Add salt at the end.

> **Tip** Do not rush the roasting; it creates the flavour.

Bhang ki Chutney

Roasted hemp-seed chutney · Makes 1 small bowl · 10 minutes

Ingredients

- 3 tbsp hemp seeds
- 1–2 green chillies
- Salt
- 1 tbsp lemon juice (galgal if available)
- 1 tbsp chopped coriander
- Pinch of roasted cumin (optional)
- 2–3 tbsp water

Method

1. Dry roast the seeds until they start to pop and smell nutty.
2. Cool, then grind with chilli, salt and water into a paste.
3. Stir in lemon juice and coriander.
4. Taste; it should be tangy and quite sharp.

> **Tip** Use only food-grade hemp seeds from a legal commercial source.

Aloo ke Gutke

Tempered potatoes · Serves 4 · 20 minutes

Ingredients

- 500 g boiled and peeled potatoes
- 2 tbsp mustard oil
- 1 tsp jakhya
- 2 dried red chillies
- ½ tsp turmeric
- 1 tsp coriander powder
- 1 tsp amchur (optional)
- Salt
- Fresh coriander

Method

1. Break the potatoes into chunks.
2. Heat oil and add jakhya and chillies.
3. Add the spices and potatoes.
4. Toss over medium heat 6 to 8 minutes until the edges crisp.
5. Finish with coriander.

> **Tip** Serve with puri, rotis or as a snack.

Mandua ki Roti

Finger-millet flatbread · Serves 4 · 20 minutes

Ingredients

- 1 cup mandua flour
- About ½ cup hot water
- Pinch of salt
- Ghee to serve

Method

1. Knead a soft dough with the hot water.
2. Pat into thick discs with oiled palms (a spoon of wheat flour makes rolling easier).
3. Cook on a hot tawa on both sides.
4. Puff briefly over the flame.
5. Serve with ghee, gahat dal, jaggery or chutney.

> **Tip** Knead while the dough is hot for a smoother texture.

Jhangore ki Kheer

Barnyard-millet pudding · Serves 4 · 45 minutes

Ingredients

- ½ cup jhangora
- 1 litre full-fat milk
- 5 tbsp jaggery or sugar
- 4 cardamom pods
- 1 tsp ghee
- Chopped nuts

Method

1. Wash and soak the millet for 30 minutes.
2. Roast it in ghee for a minute.
3. Simmer in the milk for 30 to 40 minutes, stirring, until soft and creamy.
4. Cool slightly before adding jaggery, which can split hot milk.
5. Add cardamom and nuts.

> **Tip** Serve warm.

Arsa

Rice and jaggery sweet · Makes about 12 · 1 hour plus resting

Ingredients

- 1 cup fine rice flour
- ¾ cup grated jaggery
- 3 tbsp water
- 1 tsp fennel or poppy seeds
- Ghee or oil for frying

Method

1. Melt the jaggery with the water and strain.
2. Mix into the flour for a stiff dough and rest 1 to 2 hours.
3. Shape small discs and press seeds on top.
4. Deep fry on low to medium heat until deep brown.
5. Cool completely; they firm up as they cool.

> **Tip** Take care with hot oil.

Badi ki Sabji

Sun-dried lentil nugget curry · Serves 4 · 30 minutes

Ingredients

- 1 cup badi
- 2 tbsp mustard oil
- ½ tsp turmeric
- 1 tsp coriander powder
- Chilli to taste
- Salt
- About 2 cups water
- Coriander
- Ghee

Method

1. Fry the badi in oil until golden.
2. Add the spices and water.
3. Simmer 15 to 20 minutes until soft and the gravy thickens.
4. Finish with coriander and ghee.

> **Tip** A common family version uses no onion or tomato.

Chilra

Fermented wheat crepe of Jaunsar-Bawar · Serves 4 · 20 minutes plus 3–4 hours

Ingredients

- 1 cup wheat flour
- About 1 cup water
- Pinch of salt
- Ghee and honey to serve

Method

1. Mix a smooth, pourable batter.
2. Cover and leave in a warm place for 3 to 4 hours.
3. Ladle onto a hot greased tawa and spread thin.
4. Cook until the edges lift.
5. Serve with ghee and honey.

> **Tip** The fermentation gives a faint tang.

Kumaoni Raita

Cucumber and mustard raita · Serves 4 · 10 minutes

Ingredients

- 1 large cucumber (ripe pahadi kheera if possible)
- 2 cups curd
- 1 tsp mustard seeds ground to a paste
- ½ tsp turmeric
- ½ tsp roasted cumin
- 1 green chilli
- Salt

Method

1. Grate the cucumber and squeeze lightly, keeping some juice.
2. Whisk the curd and mix in the mustard paste, turmeric and salt.
3. Rest 10 minutes.
4. Garnish with cumin and chilli.

> **Tip** Use mustard sparingly; too much turns bitter.

How to use this compendium

The atlas should distinguish published documentation, broad regional associations and claims that still require village-level verification. Food traditions vary by altitude, community, language, household and market access.

Evidence label	Meaning
Documented in research	Appears in cited academic or institutional source; local variants may exist.
Regional association	Known across a wider region, not necessarily exclusive to one district.
District research lead	A starting point for fieldwork, not a verified district speciality.
Recipe adaptation	Practical recipe based on common descriptions, not a single canonical version.

Contents

- 1. Landscape and food cultures
- 2. Traditional dish catalogue
- 3. Recipe cards
- 4. Grains, pulses, oilseeds and ingredients
- 5. Vegetables, wild foods, fruits, spices and beverages
- 6. Seasonal calendar
- 7. District-wise research matrix
- 8. Farming and preservation
- 9. Festivals, nutrition and documentation ethics
- 10. Photograph plan, sources and database schema

1. Landscape and food cultures

Uttarakhand spans high Himalayan and trans-Himalayan areas, middle hills, foothills, Bhabar and Terai plains. Altitude, rainfall, slope, irrigation, markets and community histories shape food. Garhwal and Kumaon are useful broad frames, but neither is a uniform cuisine.

Zone	Food and farming themes to investigate
High-altitude / trans-Himalayan	Buckwheat (kuttu/phapar), barley, hardy grains, potatoes, livestock products, dried foods and seasonal gathering; verify by valley/community.
Middle hills and terraces	Mandua, jhangora, gahat, bhatt, seasonal greens, mixed cropping, rain-fed fields and kitchen gardens.
Lower hills / foothills	Rice, wheat, maize, vegetables, pulses, mustard and horticulture; patterns vary with elevation.
Terai and irrigated plains	Rice, wheat, sugarcane, vegetables, dairy and diverse cuisines shaped by migration and trade.
Distinct community areas	Document Jaunsar-Bawar, Bhotiya and other community-specific traditions directly with knowledge holders.

2. Traditional dish catalogue

A peer-reviewed study documented 38 traditional cuisines in selected Uttarakhand study areas. This working catalogue is not an exhaustive statewide inventory.

Dish	Typical ingredients	Description / research note
Kafuli / Kafli	Leafy greens, thickener, spices	Greens cooked and thickened; greens vary by household.
Chainsoo / Chaisu	Roasted urad, spices, water	Roasted black gram ground and simmered; documented in study.
Phaanu / Phanu	Often gahat/horse gram	Soaked/crushed pulse cooked slowly; pulse and texture vary.
Gahat / Kulath dal	Horse gram	Pulse dish and ingredient for other preparations.
Bhatt dishes / churkani	Kala bhatt, spices	Black soybean preparation; kala bhatt appears in GI sources.
Dubuk / Dubke	Often bhatt or another pulse	Thick pulse preparation; verify local name/ingredients.
Thechwani	Crushed potato/radish/root vegetables	Root vegetables crushed rather than finely chopped.

Dish	Typical ingredients	Description / research note
Aloo ke gutke	Potato, oil, chilli, turmeric, coriander; sometimes jakhya	Spiced potato side/snack; seasoning varies.
Jholi / jhol	Curd/buttermilk, flour, spices	Tangy gravy-like dish; multiple versions exist.
Bodi / badi	Dried pulse dumplings	Cooked in curries; document pulse and drying method.
Sisunak / kandali saag	Nettle leaves, aromatics	Seasonal green; correct identification and safe handling required.
Chhachh-indu	Jhangora, buttermilk, salt, spices	Barnyard millet combined with buttermilk; recorded in study.
Raithu / raita	Curd, vegetables, spices	Yogurt accompaniment; ingredients vary.
Bhangjeera chutney	Locally named seeds, chilli, salt	Verify botanical identity; names can vary.
Urad ke pakode	Black gram batter, spices	Savory pulse fritters.
Bhang ke pakode	Locally named ingredient and batter	Verify local ingredient identity before describing.
Aarsa	Rice flour and sweetener	Festive sweet; recipes vary.
Rot	Flour and sweetener	Traditional sweet; name may refer to different foods.
Gulgula	Flour, jaggery/sugar, flavouring	Small sweet fried dumplings.
Jhangore ki kheer	Barnyard millet, milk, sweetener	Millet pudding documented as dessert.
Marsa / ramdana laddoo	Amaranth, jaggery, sometimes ghee	Grain sweet with festive associations.
Singori	Milk solids wrapped in malu leaf	Kumaoni sweet; verify leaf identity and method.
Baadi	Millet/buckwheat flour in local versions	Dense flour preparation; grain varies by locality.
Jhangora / kauni	Barnyard / foxtail millet	Boiled grain or porridge served with pulse dishes.

3. Recipe cards

Recipes are indicative adaptations. Before publishing as heritage recipes, test quantities with local cooks, record their exact process and obtain permission to attribute.

A. Aloo ke Gutke

Profile: Spiced potato side or snack.

Ingredients (indicative): 500 g potatoes; 1–2 tbsp mustard oil; cumin or jakhya; turmeric; chilli; coriander powder; salt; coriander; optional lemon.

Method: Boil potatoes until just tender, cool and chunk. Heat oil and temper seeds. Add spices briefly, then potatoes and salt. Toss until coated and lightly crisp. Finish with coriander/lemon.

Local variation: Record whether jakhya, cumin or another seed is used.

B. Chainsoo / Chaisu

Profile: Roasted black-gram dish.

Ingredients (indicative): 1 cup urad dal; mustard oil; garlic/onion optional; coriander powder; chilli; turmeric; salt; water.

Method: Dry-roast dal slowly until aromatic; cool and grind coarsely. Heat oil/aromatics, add dal and spices, then water gradually. Simmer and stir until cooked and thick.

Local variation: Study records roasted urad as a key ingredient; roast level varies.

C. Phaanu (gahat version)

Profile: Slow-cooked horse-gram preparation.

Ingredients (indicative): 1 cup gahat; oil; garlic/onion optional; coriander powder; chilli; turmeric; salt; water.

Method: Rinse and soak overnight. Drain and crush/grind. Heat oil and aromatics, add pulse paste and spices, then water. Simmer until fully cooked.

Local variation: Some versions use whole, crushed or ground pulse.

D. Kafuli-style greens

Profile: Thick leafy-green preparation.

Ingredients (indicative): Large bunch of spinach or locally used greens; garlic; chilli; salt; small amount of flour/thickener; oil.

Method: Wash and chop greens, cook with little water until soft, mash/blend. Heat oil with garlic/chilli, add greens and salt. Mix thickener with cool water before adding; simmer thoroughly.

Local variation: Greens and thickener are not fixed; wild greens need expert identification.

E. Jhangore ki Kheer

Profile: Millet pudding.

Ingredients (indicative): ½ cup jhangora; 1 litre milk; sugar/jaggery; cardamom; optional nuts.

Method: Rinse millet. Simmer in milk on low heat, stirring until creamy. Sweeten near the end and add cardamom/nuts.

Local variation: Cooking time depends on grain and soaking.

F. Bhangjeera-style chutney

Profile: Seed-based condiment.

Ingredients (indicative): 2–3 tbsp food-grade correctly identified seeds; chilli; salt; optional garlic/lemon; water.

Method: Lightly toast if local method calls for it. Grind with chilli and salt, adding water for a coarse paste. Add local optional ingredients.

Local variation: Verify botanical identity; do not infer from name alone.

G. Mandua roti

Profile: Finger-millet flatbread.

Ingredients (indicative): 1 cup mandua flour; warm water; optional salt.

Method: Add water gradually and knead. Pat into a flat round with damp palms or between sheets. Cook on hot tawa, turning carefully until done.

Local variation: Mandua dough is less elastic than wheat; record local shaping methods.

H. Gahat dal

Profile: Horse-gram dal.

Ingredients (indicative): 1 cup gahat; water; turmeric; salt; garlic; oil/ghee; chilli/coriander.

Method: Rinse and soak; pressure-cook or simmer until tender. Temper garlic/spices in oil/ghee, combine with cooked pulse and adjust thickness.

Local variation: Record fuel, cookware and cooking times.

I. Aarsa — research version

Profile: Festive sweet with local variants.

Ingredients (indicative): Rice flour; jaggery/sugar syrup; oil/ghee for frying; optional fennel.

Method: Mix flour with cooled syrup to workable consistency. Rest as local method requires, shape portions and fry gently until cooked. Cool before storage.

Local variation: Indicative only; obtain exact method and quantities from practitioner.

J. Sisunak / kandali saag

Profile: Seasonal nettle preparation.

Ingredients (indicative): Only positively identified edible nettle; aromatics; salt; oil; optional spices.

Method: A knowledgeable local cook should demonstrate safe harvesting/handling. Cook thoroughly using established local method. Never taste raw or unidentified plants.

Local variation: Safety-critical: avoid generic wild-plant identification advice.

4. Grains, pulses, oilseeds and ingredients

Local name	Common English/group	Uses and research points
Mandua / madua	Finger millet	Roti, porridge and flour blends; document landraces/calendar.
Jhangora	Barnyard millet	Boiled grain, kheer, porridge; rice alternative in some hill areas.
Kauni / koni	Foxtail millet	Boiled grain, porridge and flour.
Marsa / ramdana	Amaranth grain	Laddoo, roasted grain, flour and festive foods.
Kuttu / owa	Buckwheat	Flour dishes in some higher-altitude settings.
Phapar	Often reported as Tartary buckwheat	High-altitude flour foods; confirm species/local naming.

Local name	Common English/group	Uses and research points
Jau	Barley	Grain foods; traditional use in colder areas.
Gehun	Wheat	Roti, sweets and everyday foods.
Dhaan / satti	Rice	Boiled rice and red-rice varieties.
Makka	Maize	Flour, roasted corn, mixed cropping.
Gahat / kulath	Horse gram	Dal and phaanu-style dishes.
Urad	Black gram	Chainsoo, fritters and dried dumplings.
Kala bhatt	Black soybean/local types	Pulse dishes; GI-related record.
Rajma	Kidney beans	Cultivated in several hill valleys; record variety/place.
Masoor / chana	Lentil / chickpea	Cultivation varies by elevation and markets.
Til / sarson	Sesame / mustard	Oil, tempering and seasonal use.
Bhangjeera	Regional seed name; identity must be verified	Chutney/seasoning; never infer species from name.
Jakhya	Regional spice seed	Document local name, identification and actual use.

The peer-reviewed study lists amaranth, barley, wheat, rice, buckwheat, barnyard millet, finger millet, foxtail millet, horse gram, black gram, chickpea, kala bhatt and potato among foods grown or used locally. ICAR literature discusses traditional crops in relation to nutrition and livelihoods.

5. Vegetables, wild foods, fruits, spices and beverages

Vegetables and wild foods

• Potato, radish, pumpkin, leafy brassicas, spinach, fenugreek, beans, peas and other seasonal vegetables feature in everyday diets; availability depends on altitude and markets.

• Indigenous/minor vegetables need records for local name, verified botanical identity, edible part, season, cultivation status and seed source.

• Kandali/sisunak is documented as a seasonal green; correct identification and safe handling are essential.

• Locally named mushrooms such as chyun and guchhi are reported as monsoon foods in the cited study. Use qualified local experts; never rely on generic photos for identification.

• Wild tubers, ferns, fruits and edible plants need expert identification and conservation checks.

Fruits and horticulture

Product	Research directions
Uttarakhand Malta	Registered GI entry; document growing regions, harvest period, uses and value-added products.
Lal Chawal (red rice)	Listed as GI product on APEDA's Uttarakhand page; document variety and cooking.
Kala Bhat	GI-listed product; record seed identity, recipes and market chain.
Apple, pear, peach, plum, apricot, walnut, cherry, strawberry	Temperate horticulture in suitable hill climates; harvest varies by elevation/variety.
Mango, litchi, guava, aonla and citrus	Important in warmer zones/foothills; verify district production data.
Malta, lemon and local citrus	Juice, preserves, chutneys and market uses; document processing.
Wild berries/forest fruits	Verify species, season, edible part and conservation rules before publishing locations.

Spices and condiments

• Jakhya, bhangjeera, mustard, coriander, garlic, chilli, turmeric and local herbs can be catalogued, but verify actual use locally.

• Capture spelling variants: one ingredient may have many spellings, while one name may refer to different species.

• Separate culinary traditions from medicinal claims; traditional use is not proof of clinical efficacy.

Beverages and fermented foods

Academic overviews mention community-specific drink names such as sur/soor, chhang, jann/jan, lugdi/lugri, pakhoi/paakuyi and jhol. Do not treat these names as interchangeable. Record community, language, ingredients,

starter culture, process, alcohol content where relevant and ceremonial context with informed consent. Also catalogue chaas/mattha, fruit juices and seasonal citrus drinks.

Safety: fermented products may contain alcohol or unsafe microbes if improperly produced. Describe heritage practices without presenting unvalidated procedures as food-safety guidance.

6. Seasonal food calendar

This is a fieldwork guide, not a fixed statewide calendar. Timing shifts with elevation, aspect, rainfall and year.

Season	Food/agriculture to investigate	Questions
Spring (Mar–May)	New greens, kitchen gardens, orchard flowering, field preparation and sowing in some zones.	Which greens appear first? What is sown at this elevation?
Early monsoon (Jun–Jul)	Kharif crops, maize/millet fields, rain-fed farming, seasonal greens.	Which foods are traditionally collected safely? What is mixed-cropped?
Monsoon (Jul–Sep)	Growing crops, weeding, leafy vegetables, some reported mushrooms.	What is locally available? Which species require expert identification?
Autumn (Sep–Nov)	Harvests of millets/pulses/rice depending on locality; drying and storage.	What is harvested first? Which foods are prepared for winter/festivals?
Winter (Dec–Feb)	Stored grains/pulses, dried vegetables, preserved foods, citrus in suitable areas.	Which items are dried? How are stores protected?

For every seasonal record capture month, elevation band, district/block/village, crop stage, harvest window, local name, weather and source date. Avoid assigning one season statewide when evidence is local.

7. District-wise research matrix

The table is a research plan, not proof that a dish/crop is exclusive to the district. Verify with statistical handbooks, agriculture/horticulture departments, KVKs, FPOs and local practitioners.

District	Priority research	Verification route
Almora	Kumaoni foodways, millets, pulses, sweets, citrus/orchards	Village cooks, farmer groups, KVK, district data
Bageshwar	High/middle-hill crops, buckwheat, millets, pulses and greens	Altitude-specific surveys and seed keepers
Champawat	Hill/foothill transition, grains, pulses, fruits, markets	Block crop statistics and food practitioners
Pithoragarh	High-altitude foods, buckwheat, barley, potatoes, dried foods, Bhotiya traditions	Community-led documentation and local language names
Nainital	Hill foods, horticulture, greens and foothill/urban contrast	Separate hill blocks from foothills
Udham Singh Nagar	Terai agriculture, rice/wheat, commercial crops, dairy and vegetables	Statistics, mandi/FPO records
Dehradun	Doon valley food diversity, vegetables, citrus, rural/urban links, Jaunsar-Bawar	Block data; document Jaunsari traditions separately
Haridwar	Irrigated agriculture, wheat/rice, sugarcane, dairy, pilgrimage food economy	Crop statistics and local producers
Pauri Garhwal	Terrace farming, mandua, jhangora, gahat, bhatt, greens, preservation	Village calendars and elder interviews
Rudraprayag	Mountain agriculture, millets, pulses, seasonal/pilgrimage foodways	Altitude-stratified interviews
Tehri Garhwal	Traditional grains, terraces, stored foods and changing agriculture	Village comparisons and settlement histories
Uttarkashi	High-altitude crops, barley/buckwheat, potatoes, apples, preservation	Valley/altitude horticulture data

District profile template: district → block → village → elevation band → community/language → food/crop → local names → season → recipe/cultivation → history → current status → evidence/date → photo/audio consent.

8. Traditional farming systems

Terrace cultivation

Terraces turn slopes into cultivable benches and can reduce runoff when maintained, but walls and drainage require ongoing labour. Document construction, repair, contour alignment, water outlets, soil fertility and methods used to manage heavy rainfall.

Mixed cropping and diversity

Hill plots may combine cereals, millets, pulses, oilseeds and vegetables across fields or seasons. Record actual combinations instead of assuming one universal system. Benefits depend on species, management, weather and markets.

Rain-fed farming and water

- Document rainfall dependence, springs, streams, ponds and locally managed irrigation channels where present.
- Record sowing dates, seed rate, spacing, weeding, rotation, soil amendments and local tool names.
- Investigate spring depletion, erratic rainfall, wildlife damage, migration and market effects on crop choices.
- Do not publish sensitive water-source coordinates without permission.

Seed saving

For each variety record local name, crop species, seed/grain appearance, selection criteria, storage, sowing window, farmer-valued traits, seed keeper and whether seed is shared or sold. Obtain permission before collecting or publishing seed details.

Livestock and integrated farming

Study livestock by zone and document fodder, stall feeding/grazing, manure, household dairy and changes in herd ownership. Avoid statewide numerical claims without current official data.

Kharif and Rabi

Cycle	Broad pattern	Examples to verify
Kharif/monsoon	Sowing often aligns with monsoon; harvest varies by crop/altitude.	Rice, maize, millets, pulses and vegetables where suitable.
Rabi/winter	Sowing after monsoon; growth through cool months; harvest often in spring.	Wheat, barley, pulses and winter vegetables.
Horticulture	Perennial crops follow flowering, fruit-set and harvest calendars.	Apple, pear, stone fruits and citrus by elevation/variety.

9. Food preservation and storage

Practice	What to document	Safety/quality notes
Sun-drying	Greens, vegetables, fruit, chillies, grains/pulses; cut size, surface, weather and duration.	Dry adequately and store to limit mould.
Grain storage	Containers, granaries, moisture protection, pest management and inspection.	Record materials and safeguards; do not assume all methods equally effective.
Bodi/badi	Pulse type, batter, shaping, drying, storage and recipes.	Moisture, hygiene and complete cooking matter.
Pickling	Ingredient, salt/acid/oil method, fermentation, vessel, season and storage.	No shelf-life claims without tested safety evidence.
Fermentation	Starter, substrate, time, temperature, vessel, community use and consent.	Potential alcohol/microbial hazards; expert review needed.
Seed preservation	Selection, drying, containers, pest control, exchange and replanting.	Separate seed storage from food storage where methods differ.
Winter provisions	Foods prepared in abundance for lean seasons.	Ask how refrigeration, roads and markets changed practice.

Record preservation goal (availability, taste, storage, ritual or sale), demonstrator, date, weather, equipment, failure modes and whether the practice remains common.

10. Festivals, nutrition and responsible documentation

- Record foods for weddings, harvest events, religious observances, fairs and household ceremonies, including who prepares them and ingredient sources.
- Fasting foods and festival sweets vary by family, community, temple tradition and locality.
- Ask about food sharing, community kitchens, seasonal feasts and the meaning of particular grains/pulses.
- Do not treat one interview as universal evidence; record exact location/context.
- Obtain consent before recording sacred, restricted, private or community-owned knowledge.

Traditional grains, pulses, vegetables and seeds contribute dietary diversity, but composition varies by cultivar, soil, storage and preparation. Avoid claims that a dish cures disease. For nutrient values cite a recognised food-composition database or peer-reviewed analysis and specify food form/measurement basis.

- Useful fields: energy, protein, fibre, minerals, allergens, cooking method and serving size—only with reliable data.
- Record perceptions of foods as warming/cooling/strengthening as cultural knowledge, not clinical proof.
- Flag allergens and risks for wild foods, mushrooms, fermented products and unfamiliar seeds.

11. Photograph and media catalogue

This PDF includes image-source research links rather than embedded third-party photographs so rights can be checked. For the online atlas, create an image record for every dish, crop, tool, field practice and preservation method.

Photo field	What to capture
Primary image	Finished dish/product with natural light.
Process sequence	Ingredients → preparation → cooking/field method → final result.
Context	Kitchen, field, terrace, market, storage vessel or festival when appropriate.
Identification	Local/English name, verified botanical name, district, block, village, date.
Rights	Photographer, subject consent, licence, attribution and permitted uses.
Accuracy	Never label generic stock imagery as a specific district's dish/local variety.
Metadata	Image ID, filename, resolution, alt text, caption, source URL and licence proof.

Suggested shot list

- Mandua flour/roti, jhangora grain/kheer, kauni and marsa/ramdana.
- Gahat, urad, kala bhatt, rajma and dried bodi/badi.
- Kafuli, chainsoo, phaanu, thechwani, aloo ke gutke and bhangjeera chutney.
- Malta, red rice, black soybean and orchard products with verified locality labels.
- Terraced fields, seed selection, sowing, weeding, harvest, threshing and grain storage.
- Traditional cooking vessels, grinding stones, hearths, drying racks, storage bins and tools.

12. Source library

Use official sources for current policies/statistics, academic papers for traditional food knowledge and GI records for registered product claims. Check latest versions before publication.

S1 — Traditional food knowledge study, full text
https://pmc.ncbi.nlm.nih.gov/articles/PMC9666931/
Core source for dishes, ingredients, recipe descriptions and seasonal foods.

S2 — PubMed record for the study
https://pubmed.ncbi.nlm.nih.gov/36404996/
Abstract and publication metadata.

S3 — ICAR: Potential Role of Traditional Crops in Uttarakhand for Nutritional Security
https://epubs.icar.org.in/index.php/IndFarm/article/view/76181
Traditional crops, nutrition and livelihoods.

S4 — ICAR: Indigenous and minor vegetables of Uttarakhand Himalaya
https://epubs.icar.org.in/index.php/IndHort/article/view/105378
Indigenous vegetables and farmer-saved seed.

S5 — Indigenous land and food systems / wild foods case study
https://www.jnu.ac.in/Faculty/scgarkoti/tks/pdf/Case-study-1-Indigenous-land-and-food-systems-in-Uttarakhand_Wild-Food.pdf
Wild foods and agricultural systems; expert identification needed.

S6 — Uttarakhand Agriculture Department documents / Millet Policy
https://agriculture.uk.gov.in/documents/
Official policies and documents.

S7 — Agriculture Department sitemap/statistics navigation
https://agriculture.uk.gov.in/sitemap/
Routes to agriculture statistics and Rabi/Kharif data.

S8 — APEDA Uttarakhand GI products
https://apeda.gov.in/Uttarakhand
GI-related information including Lal Chawal and Kala Bhat.

S9 — IP India: Uttarakhand Malta Fruit GI record
https://search.ipindia.gov.in/GIRPublicSearch/Application/Details/860

Official registered GI record.

S10 — Uttarakhand horticulture publications
https://shm.uk.gov.in/publications/
Official horticulture guides and technical materials.

S11 — Horticulture & Food Processing overview
https://dohfpnew.utmv.in/about/
Broad crop overview; verify statistics separately.

S12 — ICAR Traditional Knowledge in Agriculture
https://www.icar.gov.in/index.php/en/ITK
Institutional traditional agriculture knowledge resources.

S13 — Traditional Food of Uttarakhand: overview
https://www.bpasjournals.com/life-sciences/index.php/journal/article/view/151
Additional food/beverage overview; cross-check claims.

S14 — APEDA main portal
https://apeda.gov.in/
Agricultural export and GI information.

13. Additional source discovery

• District statistical handbooks and Economic Survey publications: crop area, production and yield; cite year/page/table.

• Krishi Vigyan Kendras, GB Pant University, ICAR and state departments: crop practices, seed varieties and farmer training.

• Uttarakhand Organic Commodity Board and farmer-producer organisations: GI products, producer groups and market chain.

• Local museums, university theses, ethnobotanical studies and community publications: food names, oral histories and regional context.

• For photos use original field photography with consent or clearly licensed Creative Commons/Wikimedia Commons images; save the individual file page and licence proof.

14. Fieldwork interview template

Field	Questions / record
Record identity	ID, date, interviewer, language, district/block/village, elevation if known.
Food/crop name	Local name/pronunciation, alternate spellings, English name, botanical name if verified.
Knowledge holder	Role (cook/farmer/seed keeper/vendor/elder), consent and attribution preference.
Ingredients/seed	Variety, source, season, quantities and substitutions.
Recipe/method	Steps, tools, fuel, time, texture cues, serving and storage.
Agriculture	Sowing/harvest calendar, seed source, soil/water context and current status.
Culture	Festival, ritual, household use, social meaning and traditional preparer.
Change over time	Availability, labour, markets, migration, climate, new ingredients and youth knowledge.
Evidence	Photo/audio/video IDs, publication links, official data and page/table references.
Permissions	Consent for text, images, audio, location, commercial use, sensitive knowledge and reuse.

15. Recommended database schema

Entity	Suggested fields
Food item	food_id, local_name, alternate_names, transliteration, category, region, district links, season, evidence status
Recipe	recipe_id, food_id, ingredients, quantities, method, cook time, servings, variants, contributor, review status
Ingredient/crop	ingredient_id, local/English name, botanical name, crop group, cultivated/wild, variety, season
Place	place_id, district, block, village, elevation band, privacy-controlled coordinates, source
Agricultural practice	practice_id, crop, season, method, tools, water/soil context, knowledge holder, evidence
Preservation method	method_id, food, process, equipment, duration, storage conditions, safety review

Entity	Suggested fields
Media	media_id, linked entity, creator, caption, alt text, rights, licence, consent, source URL
Source/evidence	source_id, title, author/institution, date, URL, page/table, access date, reliability note
Interview/consent	record_id, approved attribution, consent scope, restrictions and withdrawal process

For a geographic atlas, use stable IDs and relational links instead of repeating place names in free text. PostGIS can support map browsing. Generalise or withhold precise household/wild-food locations where publication creates privacy, cultural or conservation risks.

16. Priority gaps before calling the catalogue complete

- Verify district crop/horticulture rankings with current official statistics.
- Collect village-specific recipes, especially where studies cover selected sites only.
- Confirm vernacular names and botanical identities with experts and herbarium references.
- Test recipe quantities with practitioners and record exact methods with consent.
- Add licensed photographs with provenance; never label generic imagery as locally authentic.
- Review safety language for wild plants, mushrooms, fermentation and storage.
- Create a community review and correction process for sensitive or misattributed knowledge.

Conclusion

Uttarakhand's food and agricultural heritage is a living system: grains and pulses connect to terraces and seed saving; dishes connect to seasonal availability and ecology; preservation connects to climate and household labour; food names carry language, memory and identity. A credible atlas should distinguish documented facts, regional associations and topics still needing local verification.

Recommended next step: convert this compendium into a searchable spreadsheet/database with one record per dish, crop, recipe, season, district, source and image licence.